Lecture: Numerical Methods to Solve Systems of Linear Equations

Linear Algebra, Direct Methods, Iterative Methods

Anuj Goyal
Faculty, Department of Materials Science and Metallurgical Engineering
Indian Institute of Technology Hyderabad

Linear equation to a system of linear equations

A **linear equation** in n unknowns has the form:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

- $x_1, x_2, \ldots, x_n$ — **variables (unknowns)**
- $a_1, a_2, \ldots, a_n, b$ — **constants**

A **linear system** is a set of such equations sharing the same unknowns.

System of linear equations

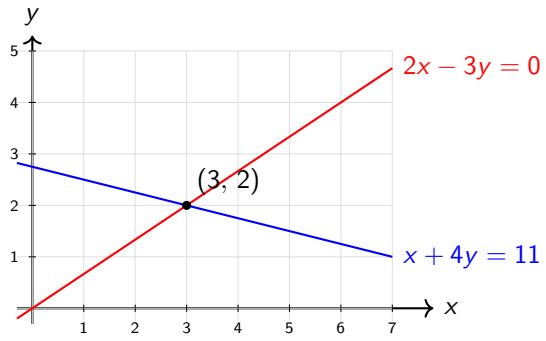
$$\begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,n-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \cdots & a_{n-1,n-1} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{n-1} \end{bmatrix}$$

Example

A materials engineer needs to produce an alloy of a specified **target composition**, starting from a few **available alloys** whose compositions are known. In what **fractions** should the available alloys be mixed so that the final alloy has exactly the target composition?

Example

Visualizing a Linear System — Row Picture



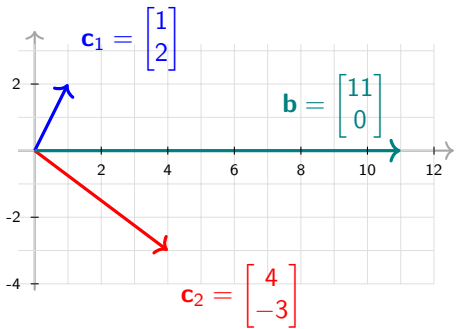
Key point

The two lines intersect at **unique point**: $(x, y) = (3, 2)$.

Visualizing a Linear System — Column Picture

Rewrite the system as a **linear combination of column vectors**:

$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 4 \\ -3 \end{bmatrix} = \begin{bmatrix} 11 \\ 0 \end{bmatrix}$$



Key point

Find x and y such that the linear combination of the *columns* of A equals $\mathbf{b}$.

Thinking About the Column Space of A

$A\mathbf{x}$ is a **linear combination of the columns of** A:

$$A\mathbf{x} = x_1\,\mathbf{c}_1 + x_2\,\mathbf{c}_2 + \cdots + x_n\,\mathbf{c}_n$$

- **The column space** $C(A)$ **= the span of the columns of A** = describes all the linear combinations of a set of vectors
- $A\mathbf{x} = \mathbf{b}$ has at least one solution vector $\mathbf{x}$ if the vector $\mathbf{b}$ is in the column space of A
- If $\mathbf{b}$ is not in $C(A)$, no combination of the columns can reach it, and the system has zero solutions

Independent and Dependent Column Vectors

Linearly Independent

$$x\begin{bmatrix}1\\2\end{bmatrix} + y\begin{bmatrix}4\\-3\end{bmatrix} = \begin{bmatrix}11\\0\end{bmatrix}$$

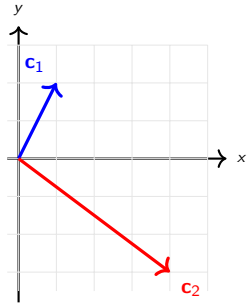
Linearly Dependent

$$x\begin{bmatrix}1\\2\end{bmatrix} + y\begin{bmatrix}2\\4\end{bmatrix} = \begin{bmatrix}1\\5\end{bmatrix}$$

Independent and Dependent Column vectors

Linearly Independent

$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 4 \\ -3 \end{bmatrix} = \begin{bmatrix} 11 \\ 0 \end{bmatrix}$$

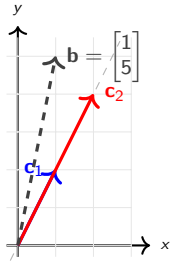


Columns span $\mathbb{R}^2$

$\Rightarrow$ **unique solution exists**

Linearly Dependent

$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$



b not in span of columns

$\Rightarrow$ **no solution**

Rank of a Matrix

$$x + 4y = 11, \quad 2x - 3y = 0$$

$$A = \begin{bmatrix} 1 & 4 \\ 2 & -3 \end{bmatrix}$$

Two **linearly independent** columns

$$\text{rank}(A) = \mathbf{2}$$

$$x + 2y = 1, \quad 2x + 4y = 5$$

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Linearly dependent columns

$$\text{rank}(A) = \mathbf{1}$$

For a square matrix ($n \times n$)

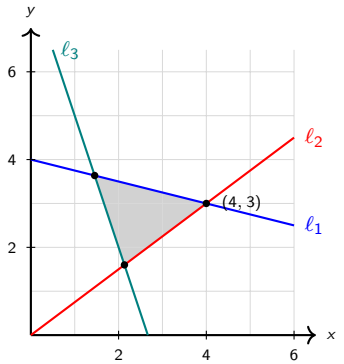
- **Full Rank ($r = n$):** $C(A) = \mathbb{R}^n \implies$ **unique solution** for every **b**.
- **Rank-Deficient ($r < n$):** $C(A) \subset \mathbb{R}^n \implies$ **0 or ∞ solutions** (depends on whether $\mathbf{b} \in C(A)$).

What happens to the nature of the solution if A is not a square Matrix?

Overdetermined System: More Equations than Unknowns

three equations, two unknowns:

$$\begin{aligned} \ell_1: &\quad x + 4y = 16 \\ \ell_2: &\quad 3x - 4y = 0 \\ \ell_3: &\quad 6x + 2y = 16 \end{aligned}$$

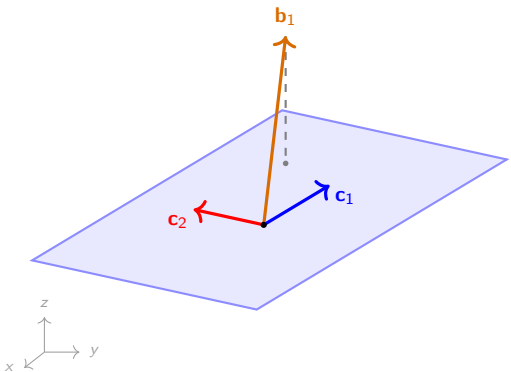


Three *different* pairwise intersections (shaded triangle)

Overdetermined System: Column Picture

Now each column is a vector in $\mathbb{R}^3$:

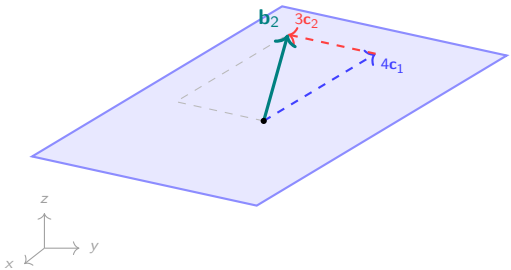
$$x \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix} + y \begin{bmatrix} 4 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} 16 \\ 0 \\ 16 \end{bmatrix}$$



Plane = column space, spanned by $\mathbf{c}_1$ and $\mathbf{c}_2$

Change b: When the Target Lies in the Column Space

$$x \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix} + y \begin{bmatrix} 4 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} 16 \\ 0 \\ 30 \end{bmatrix}$$



$\mathbf{b}_2$ lies in the column space; $\mathbf{b}_2 = 4\mathbf{c}_1 + 3\mathbf{c}_2$

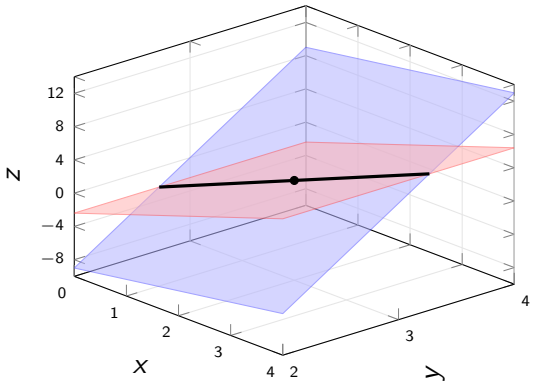
Key point

a solution **exists**, and is **unique** since $\mathbf{c}_1, \mathbf{c}_2$ are independent

Underdetermined System: Fewer Equations than Unknowns

two equations, three unknowns:

$$\begin{aligned} x + 9y - z &= 27 \\ 11x \qquad - 5z &= 12 \end{aligned}$$

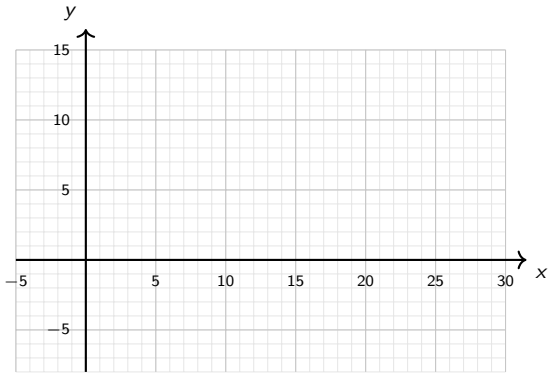


Planes $x + 9y - z = 27$ (blue) and $11x - 5z = 12$ (red); the line passes through $(2, 3, 2)$

Underdetermined System: Column Picture

$$x \begin{bmatrix} 1 \\ 11 \end{bmatrix} + y \begin{bmatrix} 9 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 27 \\ 12 \end{bmatrix}$$

Three columns, but each lives in $\mathbb{R}^2$.



Summary: Shape of the System and Its Solutions

Type	Size ($m \times n$)	Columns of A	Number of Solutions
Underdetermined ($m < n$)	2×3	3 vectors in $\mathbb{R}^2$	Infinitely many *(or none)*
Square ($m = n$)	2×2	2 vectors in $\mathbb{R}^2$	Usually exactly one
Overdetermined ($m > n$)	3×2	2 vectors in $\mathbb{R}^3$	Usually none

The Unifying Solution Diagnostic

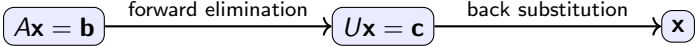
- **Existence:** A solution exists $\iff$ $\mathbf{b} \in C(A)$ (target vector lies in column space).
- **Uniqueness:** Solution is unique $\iff$ columns are independent ($N(A) = \{\mathbf{0}\}$).

Note: An underdetermined system ($m < n$) can **never** have a unique solution because at least $n - m \geq 1$ free variables always remain.

How to Solve a System of Linear Equations Numerically

- Consider a square $n \times n$ system: $A\mathbf{x} = \mathbf{b}$
- If A^{-1} exists, the exact solution is $\mathbf{x} = A^{-1}\mathbf{b}$
- But in practice, we *never* compute A^{-1}. **Why?**
- **Direct Methods:**
  - Core Foundation: **Gaussian Elimination**
  - *Coming later:* Matrix Factorizations (LU, Cholesky)

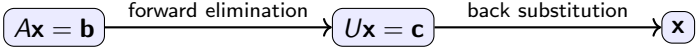
Direct Method: Gaussian Elimination



Key point

Transforming a system into upper triangular form (U) to solve for $\mathbf{x}$ is the foundational workhorse of modern scientific computing.

Gaussian Elimination: Elementary Row Operations



The forward elimination is based on three basic rules for matrix manipulation:

- Multiply row i by scalar k
- Add k times row j to row i
- Switch rows i and j

From A to U and b to c: Elimination in Column 1

$$A = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{bmatrix} \qquad \mathbf{b} = \begin{bmatrix} 19 \\ 55 \\ 50 \end{bmatrix}$$

From A to U and b to c: Elimination in Column 1

The starting matrix is A (independent columns, no row exchanges needed). **The first pivot is 2.**

$$A = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{bmatrix} \qquad \mathbf{b} = \begin{bmatrix} 19 \\ 55 \\ 50 \end{bmatrix}$$

E_{21} multiplies equation 1 by 2 and **subtracts it from equation 2**. You see the new zero.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_{21}A = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ \mathbf{0} & 5 & 6 \\ 2 & 8 & 17 \end{bmatrix} \quad E_{21}\mathbf{b} = \begin{bmatrix} 19 \\ 17 \\ 50 \end{bmatrix}$$

Completing Elimination in Column 1

The first zero is in place, but row 3 still has a 2 in column 1.

$$E_{21}A = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 0 & 5 & 6 \\ \mathbf{2} & 8 & 17 \end{bmatrix} \qquad E_{21}\mathbf{b} = \begin{bmatrix} 19 \\ 17 \\ 50 \end{bmatrix}$$

To produce another zero, *subtract row 1 from row 3* using E_{31}. This completes elimination in column 1.

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad E_{31}E_{21}A = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ \mathbf{0} & 5 & 6 \\ \mathbf{0} & 5 & 13 \end{bmatrix} \quad \begin{bmatrix} 19 \\ 17 \\ 31 \end{bmatrix}$$

Elimination in Column 2

Move now to **column 2 and row 2** (the second pivot row). The pivot is **5**, on the diagonal.

$$E_{31}E_{21}A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & \mathbf{5} & 6 \\ 0 & \mathbf{5} & 13 \end{bmatrix} \qquad \begin{bmatrix} 19 \\ 17 \\ 31 \end{bmatrix}$$

To eliminate the 5 below it, multiply row 2 by 1 and subtract from row 3.

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \quad U = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 0 & \mathbf{5} & 6 \\ 0 & \mathbf{0} & \mathbf{7} \end{bmatrix} \quad \mathbf{c} = \begin{bmatrix} 19 \\ 17 \\ 14 \end{bmatrix}$$

Pseudocode: Forward Elimination

$$\underbrace{\left[\begin{array}{ccc|c} \mathbf{2} & 3 & 4 & 19 \\ 4 & 11 & 14 & 55 \\ 2 & 8 & 17 & 50 \end{array}\right]}_{[A|\mathbf{b}]} \xrightarrow{\text{Forward Elimination}} \underbrace{\left[\begin{array}{ccc|c} \mathbf{2} & 3 & 4 & 19 \\ 0 & \mathbf{5} & 6 & 17 \\ 0 & \mathbf{0} & \mathbf{7} & \mathbf{14} \end{array}\right]}_{[U|\mathbf{c}]}$$

Forward Elimination (Python Pseudocode)

```
for k in range(0, n - 1):

    for i in range(k + 1, n):
        m = A[i, k] / A[k, k]        # Multiplier
        b[i] = b[i] - m * b[k]       # Update vector b

        for j in range(k + 1, n):
            A[i, j] = A[i, j] - m * A[k, j]
```

Forward Elimination: Arithmetic Operations Count

$$\begin{bmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0,n-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \cdots & a_{n-1,n-1} \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{n-1} \end{bmatrix} \xrightarrow{\text{elimination}} \begin{bmatrix} u_{0,0} & u_{0,1} & \cdots & u_{0,n-1} \\ 0 & u_{1,1} & \cdots & u_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{n-1,n-1} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-1} \end{bmatrix}$$

The Triangular System $U\mathbf{x} = \mathbf{c}$

$E_{32}E_{31}E_{21}A = U$ is **triangular**. The same steps applied to **b** produced the new right-hand side **c**:

$$U\mathbf{x} = \mathbf{c} \quad \text{is} \quad \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 0 & \mathbf{5} & 6 \\ 0 & 0 & \mathbf{7} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 19 \\ 17 \\ 14 \end{bmatrix}$$

Back Substitution to Solve $U\mathbf{x} = \mathbf{c}$

The pivots $2, 5, 7$ on the main diagonal of U are **not zero**

$$U\mathbf{x} = \mathbf{c} \quad \text{is} \quad \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 0 & \mathbf{5} & 6 \\ 0 & 0 & \mathbf{7} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 19 \\ 17 \\ 14 \end{bmatrix}$$

Back substitution

Back substitution	The last equation $7x_3 = 14$ gives $x_3 = 2$
Work upwards	The next equation $5x_2 + 6(2) = 17$ gives $x_2 = 1$
Upwards again	The first equation $2x_1 + 3(1) + 4(2) = 19$ gives $x_1 = 4$
Conclusion	The only solution to $U\mathbf{x} = \mathbf{c}$ is $\mathbf{x} = \begin{bmatrix} 4 & 1 & 2 \end{bmatrix}^{\mathsf{T}}$
Special note	Solving for x_1, x_2, x_3 required **dividing by the pivots** $2, 5, 7$

Pseudocode for Back Substitution

$$U\mathbf{x} = \mathbf{c} \quad \text{is} \quad \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 0 & \mathbf{5} & 6 \\ 0 & 0 & \mathbf{7} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 19 \\ 17 \\ 14 \end{bmatrix}$$

Back Substitution (Python Pseudocode)

```
for i in range(n - 1, -1, -1):
    sum_val = 0

    for j in range(i + 1, n):
        sum_val = sum_val + A[i, j] * x[j]

    x[i] = (b[i] - sum_val) / A[i, i]
```

Caution! Possible Breakdown of Elimination

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 14 \\ 2 & 8 & 17 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 3 & 4 \\ 0 & \mathbf{0} & 6 \\ 0 & 5 & 13 \end{bmatrix} = B$$

Handling Zero Pivots: Row Swapping (Pivoting)

```
if A[k, k] == 0:

    for p in range(k + 1, n):
        if A[p, k] != 0:

            A[[k, p]] = A[[p, k]]
            b[[k, p]] = b[[p, k]]
            break  # Exit pivot found
```

Caution! Possible Breakdown of Elimination

$$A^* = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 14 \\ 2 & 3 & 17 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 3 & 4 \\ 0 & \mathbf{0} & 6 \\ 0 & \mathbf{0} & 13 \end{bmatrix} = U^*$$

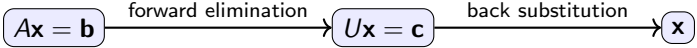
Key point

A triangular matrix U has full rank exactly when its main diagonal has no zeros

Back Substitution: Arithmetic Operations Count

$$\begin{bmatrix} u_{0,0} & u_{0,1} & \cdots & u_{0,n-1} \\ 0 & u_{1,1} & \cdots & u_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{n-1,n-1} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-1} \end{bmatrix} \xrightarrow{\text{back subst.}} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix}$$

Gaussian Elimination: Summary Operations Count



Phase	**Mult / Div**	**Add / Sub**	**Total FLOPs ($\approx$)**	**Complexity**
Forward Elimination	$\frac{N^3}{3}$	$\frac{N^3}{3}$	$\frac{2}{3}\mathbf{N^3}$	$\mathcal{O}(N^3)$
Back Substitution	$\frac{N^2}{2}$	$\frac{N^2}{2}$	$\mathbf{N^2}$	$\mathcal{O}(N^2)$
Total Direct Solve	$\frac{\mathbf{N^3}}{3} + \frac{\mathbf{N^2}}{2}$	$\frac{\mathbf{N^3}}{3} + \frac{\mathbf{N^2}}{2}$	$\frac{2}{3}\mathbf{N^3} + \mathbf{N^2}$	$\mathcal{O}(\mathbf{N^3})$

Key point

Forward elimination $\approx \frac{2N^3}{3}$ operations; **backward substitution** requires $\approx N^2$ operations

Gaussian Elimination: Algorithm Summary

```
n, x = len(b), np.zeros(len(b))

# --- FORWARD ELIMINATION WITH PIVOTING ---
for k in range(0, n - 1):
    # 1.  Find non-zero pivot & swap
    if A[k, k] == 0:
        for p in range(k + 1, n):
            if A[p, k] != 0:
                A[[k, p]], b[[k, p]] = A[[p, k]], b[[p, k]]
                break

    # 2.  Eliminate below pivot
    for i in range(k + 1, n):
        m = A[i, k] / A[k, k]
        b[i] = b[i] - m * b[k]
        for j in range(k + 1, n):
            A[i, j] = A[i, j] - m * A[k, j]

# 3.  BACK SUBSTITUTION
for i in range(n - 1, -1, -1):
    sum_val = 0
    for j in range(i + 1, n):
        sum_val = sum_val + A[i, j] * x[j]
    x[i] = (b[i] - sum_val) / A[i, i]
```